DO NOW, AO1 REVISION

Internal energy

Particle model of matter
Paper 1, Combined and Separate
12 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Define internal energy.**	The total energy stored by the particles that make up a system.	☐
2	**State how internal energy is related to the energies of the particles.**	Internal energy = total kinetic energy + total potential energy of the particles	☐
3 ★	**State which part of the internal energy changes when the temperature changes.**	The kinetic energy of the particles	☐
4	**Describe how the average kinetic energy of the particles is related to temperature.**	The greater the average kinetic energy, the higher the temperature.	☐
5 ★	**State which part of the internal energy changes when a substance changes state.**	The potential energy of the particles	☐
6	**The potential energy of the particles changes at a constant temperature. State what happens to the substance.**	It changes state.	☐
7	**State the two things that can happen when a substance is heated.**	Its temperature can increase, or it can change state.	☐
8	**Describe the change in temperature of a substance during a change of state.**	The temperature remains constant.	☐
9	**State the two things that can happen when a substance is cooled.**	Its temperature can decrease, or it can change state.	☐
10	**In which state of matter is the internal energy of a substance highest?**	Gas	☐
11	**State what heating does to the internal energy of a system.**	It increases it (the particles gain energy).	☐
12	**State three things that the temperature rise of a heated substance depends on.**	Its mass, its specific heat capacity and the energy transferred to it	☐

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